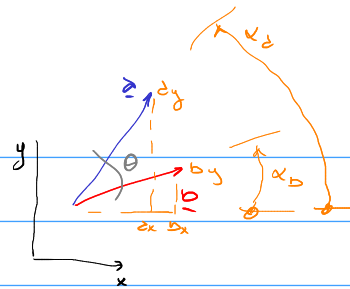


2D)

$$1) \underline{a} \cdot \underline{b} = a_x b_x + a_y b_y$$

$$2) \underline{a} \cdot \underline{b} = |\underline{a}| \cdot |\underline{b}| \cdot \cos \theta$$



$$a_x = \cos \alpha_a \cdot |\underline{a}|$$

$$a_y = \sin \alpha_a \cdot |\underline{a}|$$

$$b_x = \cos \alpha_b \cdot |\underline{b}|$$

$$b_y = \sin \alpha_b \cdot |\underline{b}|$$

$$\theta = |\alpha_a - \alpha_b| \quad 0 \leq |\alpha_a - \alpha_b| \leq \pi$$

$$\cos \theta = \cos(|\alpha_a - \alpha_b|) = \cos(\alpha_a - \alpha_b)$$

$$|\underline{v}| = \sqrt{v_x^2 + v_y^2}$$

1)

$$\underline{a} \cdot \underline{b} = \overbrace{\cos \alpha_a}^{a_x} \cdot |\underline{a}| \cdot \overbrace{\cos \alpha_b}^{b_x} \cdot |\underline{b}| + \overbrace{\sin \alpha_a}^{a_y} \cdot |\underline{a}| \cdot \overbrace{\sin \alpha_b}^{b_y} \cdot |\underline{b}| =$$

$$= (\cos \alpha_a \cdot \cos \alpha_b + \sin \alpha_a \cdot \sin \alpha_b) \cdot |\underline{a}| \cdot |\underline{b}|$$

$$= \cos(\alpha_a - \alpha_b) \cdot |\underline{a}| \cdot |\underline{b}|$$

$$= \cos \theta \cdot |\underline{a}| \cdot |\underline{b}|$$

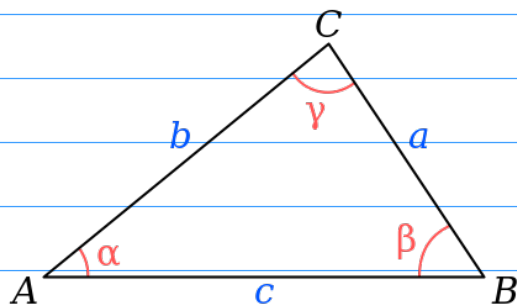
coseno de la diferencia

$$2) \underline{a} \cdot \underline{b} = |\underline{a}| \cdot |\underline{b}| \cdot \overbrace{(\cos \alpha_a \cos \alpha_b + \sin \alpha_a \sin \alpha_b)}^{\theta}$$

$$= |\underline{a}| \cdot \cos \alpha_a \cdot |\underline{b}| \cdot \cos \alpha_b + |\underline{a}| \cdot \sin \alpha_a \cdot |\underline{b}| \cdot \sin \alpha_b =$$

$$= a_x b_x + a_y b_y$$

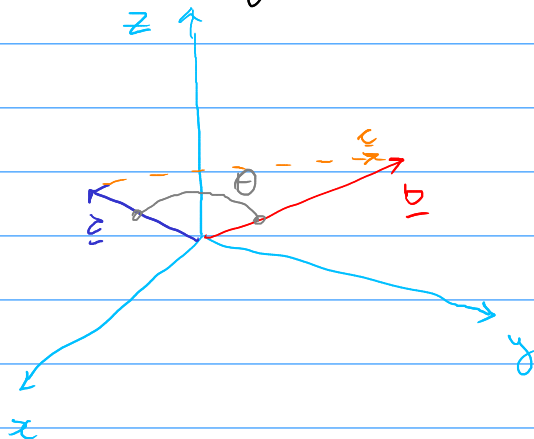
3D) usando teorema del coseno:



$$c^2 = a^2 + b^2 - 2ab \cos \gamma$$

$$ab \cos \gamma = \frac{a^2 + b^2 - c^2}{2}$$

Se lleva al problema del ángulo entre 2 vectores:



$$|\underline{a}| |\underline{b}| \cos \theta = \frac{|\underline{a}|^2 + |\underline{b}|^2 - |\underline{b} - \underline{a}|^2}{2}$$

$$= \frac{[a_x^2 + a_y^2 + a_z^2 + b_x^2 + b_y^2 + b_z^2 - (b_x - a_x)^2 - (b_y - a_y)^2 - (b_z - a_z)^2]}{2}$$

$$= \frac{[\cancel{a_x^2} + \cancel{a_y^2} + \cancel{a_z^2} + \cancel{b_x^2} + \cancel{b_y^2} + \cancel{b_z^2} - \cancel{b_x^2} + 2b_x a_x - \cancel{a_x^2} - \cancel{b_y^2} + 2b_y a_y - \cancel{a_y^2} - \cancel{b_z^2} + 2b_z a_z - \cancel{a_z^2}]}{2}$$

$$= \frac{[2b_x a_x + 2b_y a_y + 2b_z a_z]}{2} \Rightarrow$$

$$|\underline{a}| |\underline{b}| \cos \theta = a_x b_x + a_y b_y + a_z b_z$$